

Python Revision Assignment

Part 1: Data Types & Strings

- Q1. Take a user's full name as input. Print it in uppercase, lowercase, count total characters (excluding spaces), and print the first and last character.
- Q2. Input a sentence and print the number of words, number of vowels, and the reversed sentence.
- Q3. Check whether a string is a palindrome.

Part 2: Conditional Statements

- Q4. Take marks as input and print the grade (A/B/C/D/Fail).
- Q5. Check whether a year is a leap year.
- Q6. Take three numbers and print the largest.

Part 3: Loops

- Q7. Print the multiplication table of a number.
- Q8. Print all even numbers from 1 to 100.
- Q9. Find the factorial of a number.
- Q10. Print the Fibonacci series up to N terms.
- Q11. Count the digits in a number.
- Q12. Reverse a number.

Part 4: Functions

- Q13. Create a function that returns the square of a number.
- Q14. Create a function that checks whether a number is prime.
- Q15. Create a function that accepts positional and keyword arguments.

Part 5: Variable Scope

- Q16. Demonstrate global and local variables.
- Q17. Modify a global variable using the global keyword.

Part 6: Exception Handling

- Q18. Handle ZeroDivisionError and ValueError while dividing two numbers.
- Q19. Open a non-existent file and handle FileNotFoundError.
- Q20. Create a custom exception using raise.

Part 7: File Handling

- Q21. Create 'student.txt' and write your details.
- Q22. Read the file contents.
- Q23. Append a new line to the file.

- Q24. Count total lines, words, and characters.

Part 8: Python os Module

- Q25. Create a folder named 'PythonPractice'.
- Q26. Create three empty text files inside it.
- Q27. Rename one file.
- Q28. List all files in the folder.
- Q29. Delete all files and remove the folder.

Mini Project

- Build a Student Record Management System with options to Add, View, Search, Delete students, and Exit. Use functions, loops, file handling, and exception handling.

Challenge Projects

- 1. Number Guessing Game
- 2. Password Strength Checker
- 3. ATM Simulation
- 4. Rock Paper Scissors
- 5. Quiz Game
- 6. To-Do List using File Handling